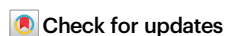


Charge-domain content addressable memory based on ferroelectric capacitive memory for reliable and energy-efficient one-shot learning

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Non-volatile content addressable memories (NV-CAMs) accelerate memory augmented neural networks (MANNs) for brain-like efficient learning from a few examples or even one example. However, most existing NV-CAMs operate in current domain, posing challenges in reliable, low-power, and sensing-friendly Hamming distance (HD) computation. To address these challenges, this work proposes transferring the computation to charge domain using ferroelectric capacitive memory (FCM). For the first time, a charge-domain 2FCM CAM based on the inversion-type FCM is reported. By storing data as device capacitance, this CAM structure directly outputs HD as linear multi-level voltages, enabling simplified sensing processes and reduced peripheral costs. Its differential nature further exhibits immunity to device variation, ensuring accuracy in the computation of long data vectors. Parallel 16-bit HD computation using a fabricated 16×16 2FCM CAM array is experimentally demonstrated with record performance at array level, evidencing the superiority of charge-domain computation and showcasing tremendous potential for in-memory-search applications.

Deep neural networks (DNNs)¹ demonstrate remarkable capabilities across diverse applications, but at the expense of massive data and vast computational resources. Handling unseen data with a limited number of examples challenges DNNs, and re-training all the network parameters is often required for new instances. Meanwhile, many existing hardware resources are struggling to meet the surging demand for computing power, especially in Internet of Things (IoT) devices. In comparison, a biological brain acquires knowledge by learning and generalizing from only a few examples (or even one example) with a fast rate and much lower power consumption. This

disparity motivates the search for more efficient brain-inspired algorithms.

In recent years, memory-augmented neural networks (MANNs)² have emerged as a promising solution, which is capable of one-shot or few-shot learning by utilizing an associative memory to store and retrieve the extracted features. However, in conventional von-Neumann architectures³, the data search operation in MANNs involves extensive data transfer and distance calculation, resulting in long latency and high energy consumption. There have been several efforts to accelerate MANNs by embracing the in-memory computing

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paradigm, where content-addressable memories (CAMs)^{4–6} play decisive roles (Fig. 1a). By employing CAMs, an input query feature vector can be parallelly compared with multiple stored vectors, and the degree of mismatch, i.e., hamming distance (HD), between them can be directly calculated within the memory. By converting the extracted features into binary vectors via transformation functions such as the locality sensitive hashing (LSH) function⁷, MANNs can be significantly

accelerated with CAMs, attributable to the high parallelism and minimized data movement between computing units and associative memory⁴.

Figure 1b compares several typical circuit designs based on different electron devices to realize the function of CAM. Conventional CAM utilizes the complementary-metal-oxide-semiconductor (CMOS) static random-access memory (SRAM) as the memory cell⁸. While the

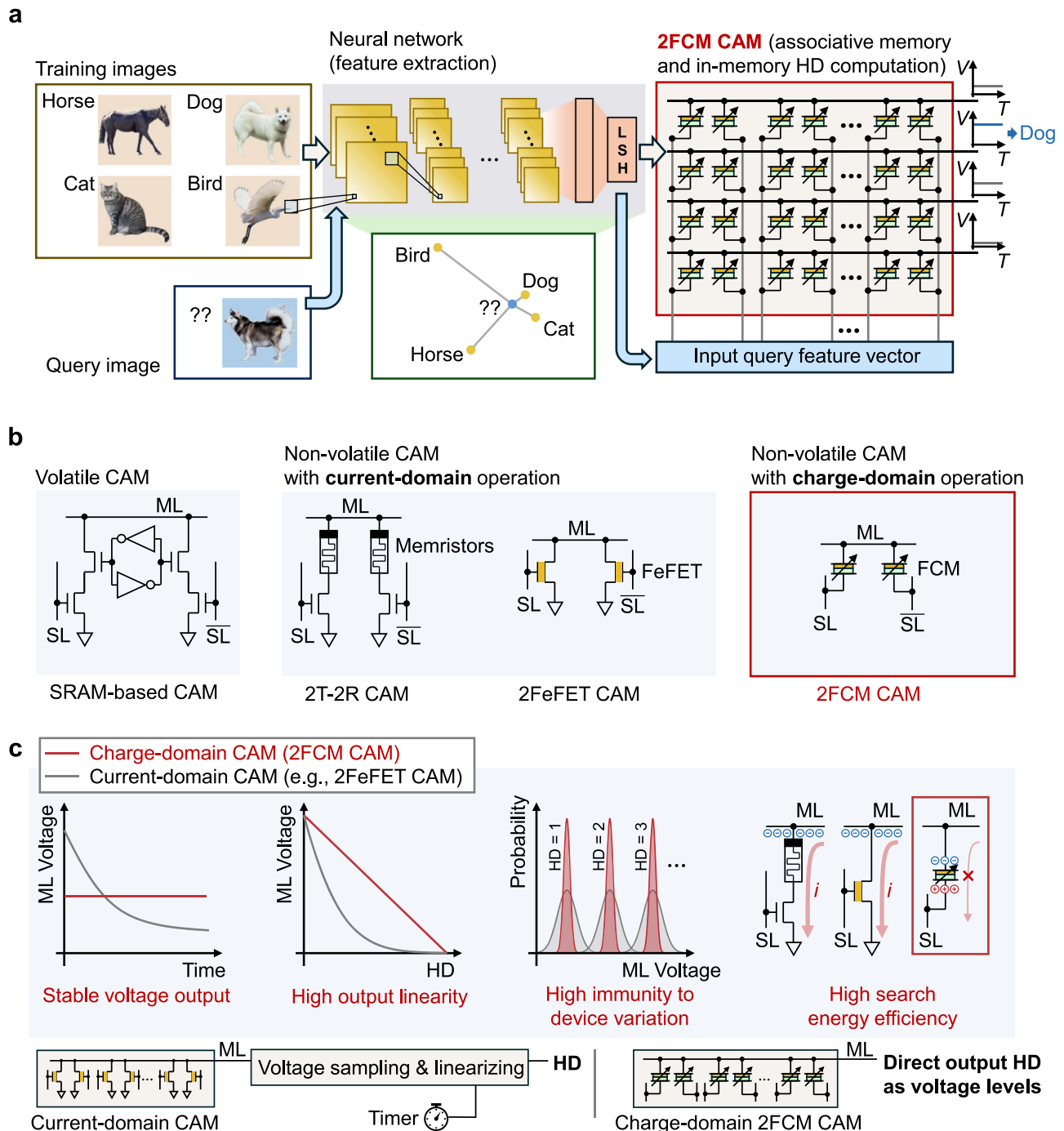


Fig. 1 | MANN accelerated by CAM. **a** One-shot learning task implemented by MANN with CAM-based associative memory. In the MANN structure, a convolutional neural network extracts the features of the examples and maps them to the binary vectors through transformation functions such as Locality-Sensitive Hashing (LSH). The obtained binary vectors from the training datasets can be stored in the NV-CAM arrays (e.g., the proposed 2FCM CAM array) as the stored feature vectors. The classification of the unseen query example is conducted by finding the vector with the closest distance from the stored feature vectors to the unseen one. By

inputting the query feature vectors to the NV-CAM array, the HD to each stored vector can be parallelly calculated in the memory to accelerate the classification process. **b** Comparison of the structures of 3 typical current-domain CAMs and the proposed charge-domain 2FCM CAM. 2FCM CAM operates in the charge domain without direct leakage paths from the match line (ML) to the ground. **c** Illustration of the advantages provided by the charge-domain 2FCM CAM over the current-domain CAMs, including stable voltage output, high output linearity, high immunity to device variation, and high search energy efficiency.

SRAM-based CAMs are mature, reliable, and fast, they suffer from volatility and are unsuitable for long-term data storage in IoT devices due to the costly standby power⁹. Moreover, its large cell size impedes further improvement of the integration density. In the past decade, the rapid development of non-volatile memories (NVMs) has sparked a new wave of innovations in the research of CAMs. Various NVM technologies have been employed to build the non-volatile CAMs (NV-CAMs) including flash memory^{10,11}, resistive random-access memory (RRAM)^{6,12,13}, phase change memory (PCM)¹⁴, magnetic tunnel junction (MTJ)^{15,16}, ferroelectric field-effect transistor (FeFET)^{4,17–19}, ferroelectric tunnel field-effect transistor (FeTFET)²⁰, etc. Compared with SRAM-based CAMs, NVM-based CAMs effectively eliminate the standby leakage power by using NVM characteristics and dramatically enhance area efficiency with smaller footprints of the memory cells.

However, several challenges still exist, hindering further advancement of these NV-CAMs. For the NV-CAMs based on resistive devices (i.e., memristors) including MTJ, PCM, and RRAM, the write current can be high due to the current-driven write mechanism, which results in substantial write energy consumption and necessitates large access transistors to drive the write operation²¹. Besides, the low ON/OFF ratio of some resistive devices^{15,16} also causes high sensing complexity and hardware costs. In contrast, the FeFET-based NV-CAMs exhibit a high ON/OFF ratio and enjoy much lower write energy due to the field-driven write mechanism⁴. Nevertheless, limitations inherent in the current-domain computing design are still unsolved (Fig. 1c). First, the NVM devices in the current-domain CAMs are used to control the resistance from the match line (ML) to the ground. To avoid the large current, HD computation is performed by inputting the search data through the search lines (SLs) and tracking the voltage drop of the pre-charged MLs^{4,6,10–20}. To sense this dynamic voltage output, precise timing controls are indispensable, leading to high peripheral sensing overheads. Second, such a sensing scheme, in theory, is unable to provide the results as linear voltage levels. The non-linearity further complicates accurate sensing and HD computation in various applications. Third, the device-to-device variation in NVMs is often challenging to control, especially for aggressively down-scaled devices^{22–25}. The current-domain sensing scheme is highly susceptible to the intrinsic device variation, severely limiting the maximum distinguishable output levels in the CAM arrays and hence degrading the accuracy of the MANNs. Fourth, NV-CAMs still suffer from high power dissipation due to inefficient search operations. It should be emphasized that the optimization of search power is even more important to realize low-power CAMs when deployed in IoT devices, where search operations are repeatedly executed with continuous input query feature vectors, whereas the write operation is only needed for initially writing the pre-trained models into the CAM arrays. In addition, there has also been an effort²⁶ to utilize metal-ferroelectric-metal (MFM) capacitors for the CAM implementation, where one read transistor and two MFM capacitors (1T2C) were used to construct the CAM cell. However, this CAM structure was still designed within the current-domain framework. Moreover, the quasi-nondestructive read out scheme of MFM capacitors in the 1T2C CAM design²⁶ raises concerns regarding read disturbance, which makes this design unsuitable for search-intensive applications.

To tackle these challenges, we demonstrate a completely new CAM space using ferroelectric capacitive memory (FCM)^{27,28}. This emerging NVM device with unique capacitive operation mechanism enables the charge-domain CAM operation in a compact design. The charge-domain operation has great potential to overcome the above insurmountable challenges of the current-domain CAMs based on resistive devices or FeFETs. In FCM, the stored data is represented by the different capacitance states instead of the resistance difference (i.e., junction resistance or channel resistance of the conventional NVMs). By leveraging the HfO₂-based ferroelectric materials to modulate the device capacitance, FCM also enjoys ultra-low write power, excellent CMOS compatibility, scalability, and data retention.

Recently, we presented an inversion-type FCM^{27,28} where, with the use of ferroelectric polarization to control the silicon inversion and depletion, the device further exhibits a high capacitance ratio, providing an excellent opportunity to realize NV-CAMs with reliable and sensing-friendly multi-level outputs.

By exploiting the distinctive FCM characteristics, in this article, we pioneer the first charge-domain CAMs based on the inversion-type FCM, i.e., 2FCM CAM, featuring stable voltage output, ultra-high output linearity, excellent variation immunity, and high search energy efficiency (Fig. 1c). Different from the current-domain CAMs with dynamic sensing, our innovative charge-domain 2FCM CAM delivers stable and linear voltage output scaled by HD, which, coupled with the sensing margin enlarged by inversion-type FCM, enables simple and reliable sensing of multi-level NV-CAM. Furthermore, the new operation mode is based on a differential circuit structure that effectively alleviates errors introduced by device variations. Besides, operating in the charge domain, 2FCM CAM eliminates the energy-hungry pre-charge operation in the current-domain CAMs, offering remarkably high energy efficiency for search operations. Compared to previous efforts in realizing charge-domain computing¹⁹, which involved the additional CMOS capacitors within the cells, the intrinsic capacitive nature of FCM guarantees a seamless transition of CAM operations from current-domain to charge-domain, maintaining the compact cell structure and high integration density.

To demonstrate the proposed CAM structure and its superiorities, a 16 × 16 2FCM CAM array was fabricated by integrating 32 × 16 high-performance inversion-type FCM devices on the silicon-on-insulator (SOI) platform. Ultra-high linearity ($R^2 = 0.9983$) with the average error of merely 0.126 least significant bit (LSB) for 16-level voltage output was confirmed by array-level measurement, which further supports the successful demonstration of 16-channel parallel HD computation for 16-bit data vectors. Based on the exceptional performance at the array level, we have applied the proposed 2FCM CAM as the associative memory of MANN and performed the one-shot and few-shot learning tasks for benchmarking. Comprehensive evaluations reveal that the proposed 2FCM CAM can achieve a high accuracy of 97.5% on the Omniglot dataset based on the measured results (only 0.6% accuracy loss compared with 98.1% software accuracy using LSH) and an ultra-low search energy consumption of only 0.005 fJ/bit (more than 40× lower than the state-of-the-art current-domain CAMs). Furthermore, the MANN accuracy on the Omniglot dataset remains high (~90%) even under severe device-to-device variations with device down-scaling, showcasing a ~1.7× accuracy improvement compared with current-domain NV-CAM and paving the way for the integration of large-scale and high-density CAM array. Such remarkable results open a completely new era of CAM exploration and present the enormous potential of the proposed 2FCM CAM as a competitive solution for accurate and energy-efficient one-shot and few-shot learning as well as other in-memory-search applications.

Results

2FCM CAM cell with charge-domain search operation

The proposed charge-domain CAM design is essentially based on the capacitive memory devices^{27–32} (or memcapacitors) to store data with programmable capacitance, and the inversion-type FCM^{27,28} can perfectly meet this need for friendly sensing with a large capacitance ratio modulated by ferroelectric polarization. As shown in Fig. 2a, the inversion-type FCM device for this demonstration features a metal-ferroelectric-semiconductor (MFS) structure, where a 10 nm Al-doped HfO₂ (HAO) ferroelectric thin film is sandwiched between the W top electrode (TE) and the lightly doped p-type (p) silicon. A heavily doped n-type (n⁺) region is introduced with a small overlap at the edge of the TE to provide carriers for fast ferroelectric switching. Figure 2b, c elucidates the capacitance modulation mechanism of the inversion-type FCM. When sufficient voltages are applied between the TE and

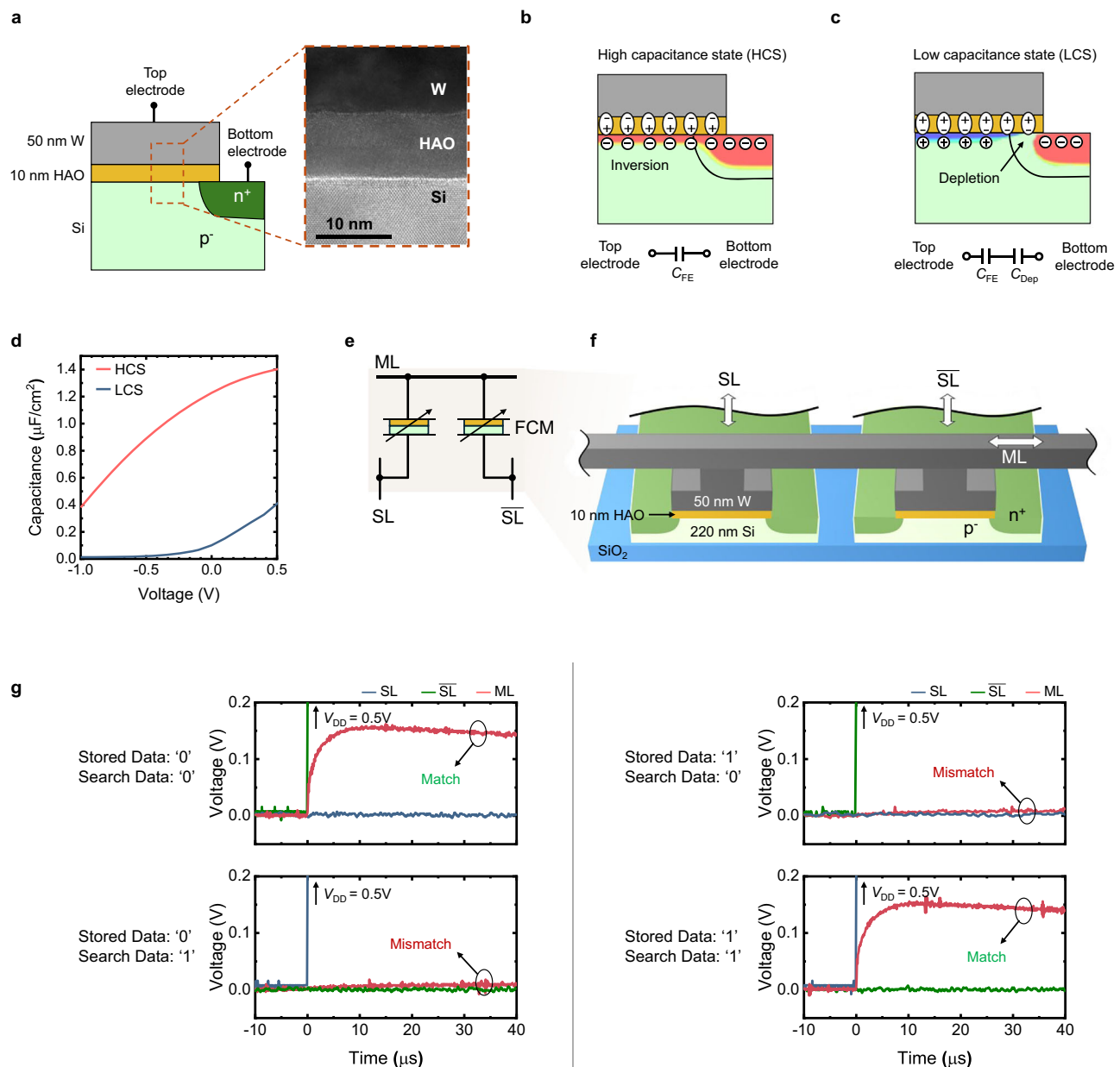


Fig. 2 | 2FCM CAM cell basics. **a** Device structure of the inversion-type FCM and high-resolution transmission electron microscope cross-section image of the W/HAO/Si stack. **b**, **c** Illustration of the carrier distribution in the Si layer for HCS and LCS, respectively. In these two memory states, the Si layer is in inversion or depletion, respectively, leading to the different capacitance measured from the two electrodes. The two memory states can be switched by applying voltage across the two electrodes with non-volatility. **d** C-V characteristics of the fabricated FCM in the two memory states. The measurement was performed under a small signal with the root-mean-square voltage of 50 mV and 100 kHz frequency. **e** Structure of a 2FCM CAM cell. The 2 FCM devices storing complementary states are connected to

represent 1-bit data in a CAM cell. **f** Structure of the fabricated 2FCM CAM cell on the SOI substrate. With the SOI substrate, the FCM devices in a row can be effectively isolated from each other by etching through the Si layer. The heavily doped n⁺ region connects the FCM devices in the same column and works as the SL or $\overline{\text{SL}}$. The ML is formed on top of an isolation layer and connected to the TEs of the FCM devices through the via holes. **g** Demonstration of the charge-domain search operation of a fabricated 2FCM CAM cell. The measured voltage waveforms of the SL/ $\overline{\text{SL}}$ (input) and the ML (output) are plotted to show the four possible cases in a 2FCM CAM cell. 0.5 V V_{DD} was applied. The search result can be correctly determined by reading the ML voltage.

bottom electrode (BE), the polarization of the HAO layer can be switched, thus writing the device into high capacitance state (HCS) or low capacitance state (LCS). In HCS, the silicon surface enters inversion due to ferroelectric polarization, and the inversion charge connects BE (n⁺ region) with the area under TE. Therefore, the capacitance between the two electrodes becomes high and is close to the capacitance of a MFM capacitor with the same area of TE (C_{FE}). In comparison, polarization in LCS lifts the surface potential of silicon, leading to a depletion region at the p-n junction. In this case, the capacitance is low and

approximately equivalent to the C_{FE} and a depletion capacitance (C_{Dep}) in series. The two capacitance states can be retained by the polarization without external voltages for non-volatile long-term data storage. The fabrication process and the polarization-voltage (P - V) characteristics of our fabricated FCM devices are provided in Supplementary Fig. 1. Figure 2d shows the measured capacitance-voltage (C - V) curves of the devices, where the capacitance is measured by the high-frequency AC small signal under varying DC voltage bias. A clear hysteresis window can be observed in the curves with a high capacitance

ratio under a negative voltage bias. The device also demonstrates excellent data retention of 10 years and decent endurance of $\sim 10^6$ cycles (Supplementary Fig. 2).

As shown in Fig. 2e, the proposed 2FCM CAM cell consists of two FCM devices. Their TEs are connected together as ML, and the BEs are connected to SL and $\overline{\text{SL}}$, respectively. In this way, these two FCM devices are connected in series between SL and $\overline{\text{SL}}$, forming a voltage divider with ML as the output. In each cell, the two FCM devices are written to complementary capacitance states. A charge-domain search operation is performed by inputting the search data as the voltages (V_{DD} and 0 V) through SL and $\overline{\text{SL}}$. When V_{DD} is applied to the BE of the FCM device in HCS and the BE of the other device in LCS is connected to ground, the ML voltage (V_{ML}) can be calculated as:

$$V_{\text{ML}} = \frac{C_{\text{HCS}}}{C_{\text{HCS}} + C_{\text{LCS}}} V_{\text{DD}} \quad (1)$$

where C_{HCS} and C_{LCS} are the capacitance values of the FCM devices in HCS and LCS, respectively. This search result represented by V_{ML} is defined as a 'match'. On the contrary, when V_{DD} is applied to the BE of the device in LCS and the BE of the other device in HCS is grounded, the ML outputs a lower voltage representing the 'mismatch' search result as:

$$V_{\text{ML}} = \frac{C_{\text{LCS}}}{C_{\text{HCS}} + C_{\text{LCS}}} V_{\text{DD}} \quad (2)$$

For an FCM device with a high capacitance ratio ($C_{\text{HCS}}/C_{\text{LCS}} \gg 1$), V_{ML} approaches V_{DD} and 0 V for 'match' and 'mismatch', respectively, which offers a large sensing margin to distinguish the different search results (Supplementary Fig. 3).

To fabricate 2FCM CAM, we integrated the inversion-type FCM devices on the SOI substrate. Figure 2f depicts the structure of a 2FCM CAM cell. In this design, the two adjacent devices are isolated by etching through the silicon layer. The n^+ regions of the FCM devices form SLs and $\overline{\text{SL}}$ s, and their TEs are connected with a metal line as ML. Figure 2g presents a demonstration of the charge-domain search operation of the fabricated cell. With a 0.5 V V_{DD} , a -0.15 V V_{ML} could be clearly detected when the input search data matches the store data, while negligible V_{ML} could be observed in the 'mismatch' case. It should be pointed out that the relatively low V_{ML} of the matched cell compared to V_{DD} and its decreasing trend were caused by the input impedance of our measurement setup (Supplementary Figs. 4–6). By using an optimized sensing circuit integrated on chip, the sensing margin can be enlarged with a faster transition time.

2FCM CAM array with highly linear HD computation

The charge-domain search operation can be easily extended to the array level to perform HD computation on a 2FCM CAM array with multi-bit input data. Figure 3a shows the structure of a CAM row with N cells. Under an input voltage of V_{DD} for all cells, V_{ML} can be formulated as:

$$V_{\text{ML}} = \frac{(N - n_{\text{mis}})C_{\text{HCS}} + n_{\text{mis}}C_{\text{LCS}}}{N(C_{\text{HCS}} + C_{\text{LCS}})} V_{\text{DD}} \quad (3)$$

which is linearly scaled with the number of mismatched cells (equal to HD), n_{mis} . For the FCM devices with a high capacitance ratio, V_{ML} can be approximated as:

$$V_{\text{ML}} \approx \frac{N - n_{\text{mis}}}{N} V_{\text{DD}} \quad (4)$$

The high linearity and time-independent feature largely simplify the sensing process and required sensing peripherals, enabling much

lower hardware cost (Supplementary Figs. 7 and 28). This is in marked contrast to the current-domain CAM arrays based on resistive devices or FeFETs, where the output voltage decays exponentially over time in each search operation and is theoretically nonlinear when sampled at a specific time point. Such nonlinearity narrows the sensing margin for the highly mismatched results, making it extremely difficult to obtain accurate HD when the number of bits increases (Supplementary Figs. 9 and 10).

To demonstrate the multi-level linear output of the proposed 2FCM CAM, a measurement shown in Fig. 3b was conducted with a row of fabricated 16-bit 2FCM CAM. Before the test, all the cells in the CAM row were written to state '1'. Next, seventeen 16-bit binary vectors from all '0's to all '1's were input to the CAM in sequence as the input query vectors. The V_{ML} was recorded as the output (Supplementary Fig. 15), whose amplitudes in each search operation are plotted in Fig. 3c. When the number of mismatched cells increases from 0 to 16, V_{ML} decreases linearly, consistent with the trend predicted by the above formulas. The voltage amplitudes can be fitted by a linear function with a high R^2 of over 0.9983. With this ultra-high linearity, the output voltage levels are evenly separated, making all the search results clearly distinguishable. This is also supported by the large sensing margin enabled by the high capacitance ratio of the inversion-type FCM.

In addition to the one-dimensional CAM row, we further extended the demonstration to the 16×16 2FCM CAM array to perform parallel HD computation. The connection and the scanning electron microscope image of the fabricated array are shown in Fig. 3d, e respectively. With the shared SLs and $\overline{\text{SL}}$ s, the CAM array is capable of parallelly comparing the input query vector with multiple stored data vectors and directly outputting the matching degree (i.e., the number of matched cells) as the multi-channel voltage levels. The 16-channel parallel HD computation was demonstrated through a test shown in Fig. 3f. Sixteen different 16-bit data vectors were first written into the CAM with the write scheme given in Supplementary Fig. 13a. The stored data were labeled as Data #1 to Data #16. After that, the same sets of data were used as the query vectors and input to the CAM via SLs and $\overline{\text{SL}}$ s in sequence. The voltage waveforms in the 16 MLs were recorded (Supplementary Fig. 19) to obtain the output of each search operation. The measurement results are displayed in Fig. 3g, where the color in each pixel indicates the matching degree calculated by the fabricated 2FCM CAM array for each two sets of data. The results show minor difference when compared with the ones processed by the software (Fig. 3h), evidencing a high accuracy of the HD computation realized by the 2FCM CAM array. We would like to highlight that, to obtain the results presented in Fig. 3g, we only applied a linear function to normalize V_{ML} . Thanks to the ultra-high output linearity provided by the 2FCM CAM structure, the simple mapping of the output voltage to the matching degree is sufficient to obtain a high accuracy for 16-bit data (Fig. 3i). Considering our experiment conditions, we believe the negligible error in the results are mainly caused by the device variation introduced during the fabrication processes and the measurement error due to the self-built driver circuit. Lower error rate can be achieved with optimized processes and low-noise sensing circuits.

Immunity to device variation

With the differential circuit structure, the ML voltage of 2FCM CAM follows a rational function, which is determined by the ratio between the capacitance values instead of the exponential decay in the current-domain CAMs. Such an operation scheme not only provides stable output voltages with high linearity but also offers immunity to the device variation (as modeled in Supplementary Figs. 7 and 8). For a fair comparison, we evaluate the current-domain 2FeFET CAM (Fig. 4a) and the charge-domain 2FCM CAM (Fig. 4b) under the same device sizes and device variations, which is considered by using the identical standard deviations, $\sigma(V_{\text{TH}})$ and $\sigma(V_{\text{FCM}})$, for the FeFET threshold voltage (V_{TH}) and the FCM internal voltage bias (V_{FCM}), respectively.

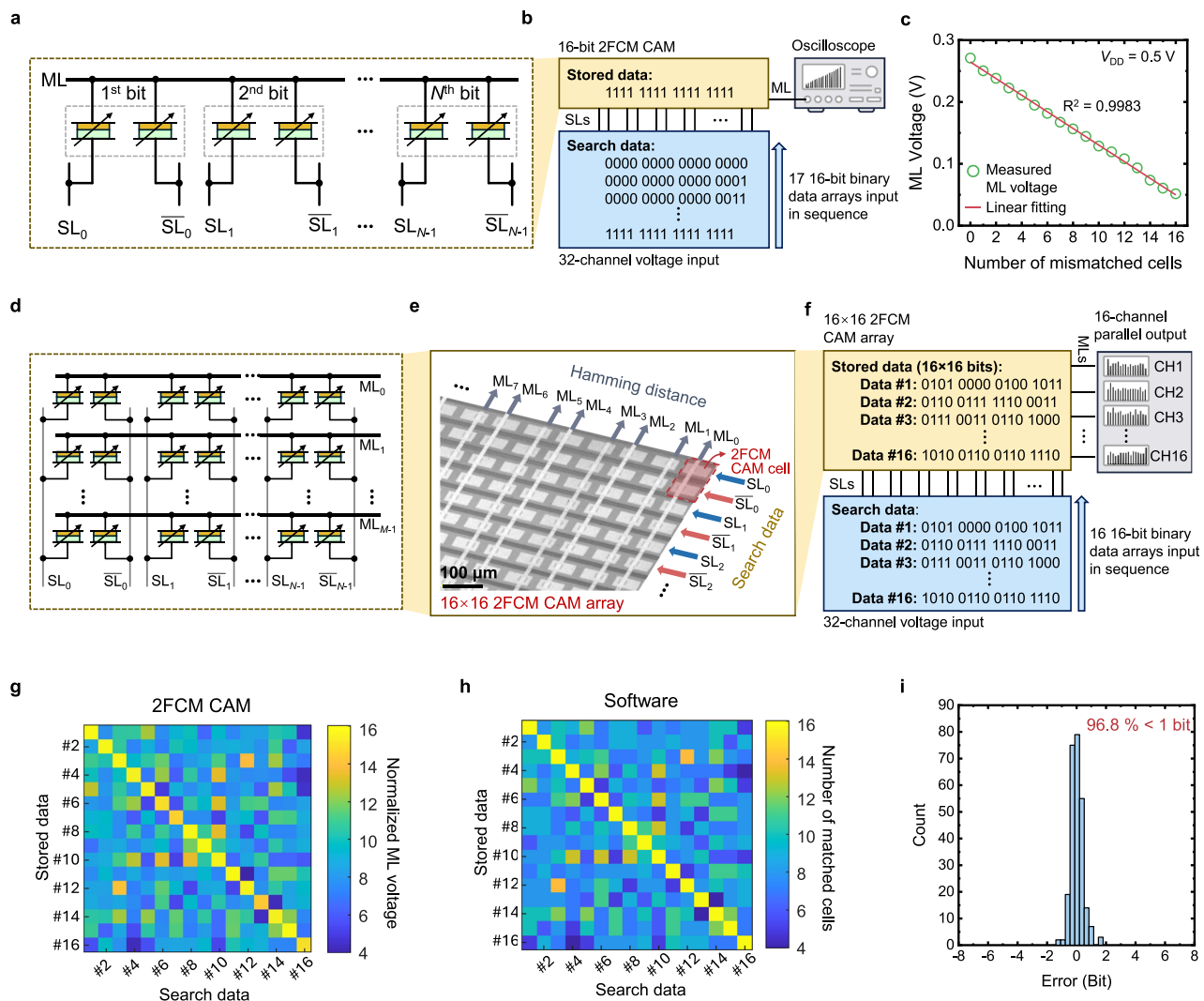


Fig. 3 | Demonstration of the 16 × 16 2FCM CAM array operation. **a** Structure of a N -bit 2FCM CAM row. **b** Illustration of the measurement setup for the 16-bit 2FCM CAM row. All 2FCM CAM cells in the row were written to store bit '1'. Seventeen 16-bit binary query vectors with the increasing number of '1' were input to the CAM row in sequence. V_{ML} was recorded by an oscilloscope. **c** Measured V_{ML} amplitudes for each search operation. The V_{ML} amplitudes can be fitted by a linear function with an R^2 of 0.9983, showing the ultra-high linearity of the output voltage. **d** Structure of a $M \times N$ 2FCM CAM array (M channels and N bits in each channel). **e** Scanning electron microscope image of the fabricated 16 × 16 2FCM CAM array and the connection scheme. **f** Measurement setup for the demonstration of 16-channel parallel HD computation. Sixteen 16-bit binary data vectors (Data #1 to Data #16) were first written into the fabricated 16 × 16 2FCM CAM array. The V/3 write scheme was implemented to perform the array-level memory write operation

for lower write disturbance among the FCM devices (Supplementary Figs. 13 and 14). Next, the same set of data (Data #1 to Data #16) were used as the search data and input to the CAM array in sequence. The 16-channel voltage waveforms at the MLs were recorded as the output. **g** Measured output voltages of the 2FCM CAM array. Each pixel in the image corresponds to a search operation. The output voltages were normalized to better indicate the number of matched cells and represented by the pseudo colors. The high values along the diagonal line show the results of complete matching when comparing a data with itself. **h** Matching degree computed by software as a reference. **i** Distribution of the bit error for all the results computed by the CAM array. The tight distribution of the error close to 0 shows high computational accuracy. In 96.8% of the search operations, HDs can be correctly obtained with <1 bit errors. The above tests were performed on a self-built measurement platform (Supplementary Figs. 11 and 12).

According to the measured device variation data from a representative FeFET array²², 0.09 V and 0.18 V were assumed as the standard deviations for 500 nm × 500 nm and 240 nm × 34 nm devices (FeFET or FCM), respectively. Figure 4c, d compares the output V_{ML} distribution of the two CAM structures with 500 nm × 500 nm device size. The proposed 2FCM CAM enjoys ~5.1× lower output variation compared to 2FeFET CAM. The output variation is defined by $\sigma_{\text{norm}}(V_{ML}) = \sigma(V_{ML})/V_{DD}$, where $\sigma(V_{ML})$ is the standard deviation of V_{ML} . When the device size shrinks to 240 nm × 34 nm, variation increases and the outputs of the CAMs are displayed in Fig. 4e, f. Under the larger device variation, 2FCM CAM can still maintain only 0.27 LSB $\sigma_{\text{norm}}(V_{ML})$, which is ~6.1× lower than 2FeFET CAM. The notable variation immunity of 2FCM

CAM, combined with its stable and linear output voltage, enables much more reliable and accurate HD computation compared to the current-domain CAMs, which ensures better scalability of the CAM array to support longer search vectors (Supplementary Fig. 8). This can not only achieve higher accuracy in the MANN tasks, but also cover more in-memory-search applications^{33–36} requiring the accurate HD computation for long vectors. Furthermore, 3D capacitor structure similar to the advanced dynamic random-access memory technology (e.g., 3D pillar capacitors^{37,38}) can be applied in 2FCM CAM to realize a larger effective device area for a lower device variation under the same footprint (Supplementary Figs. 20–24, 32–34, and Supplementary Table 1).

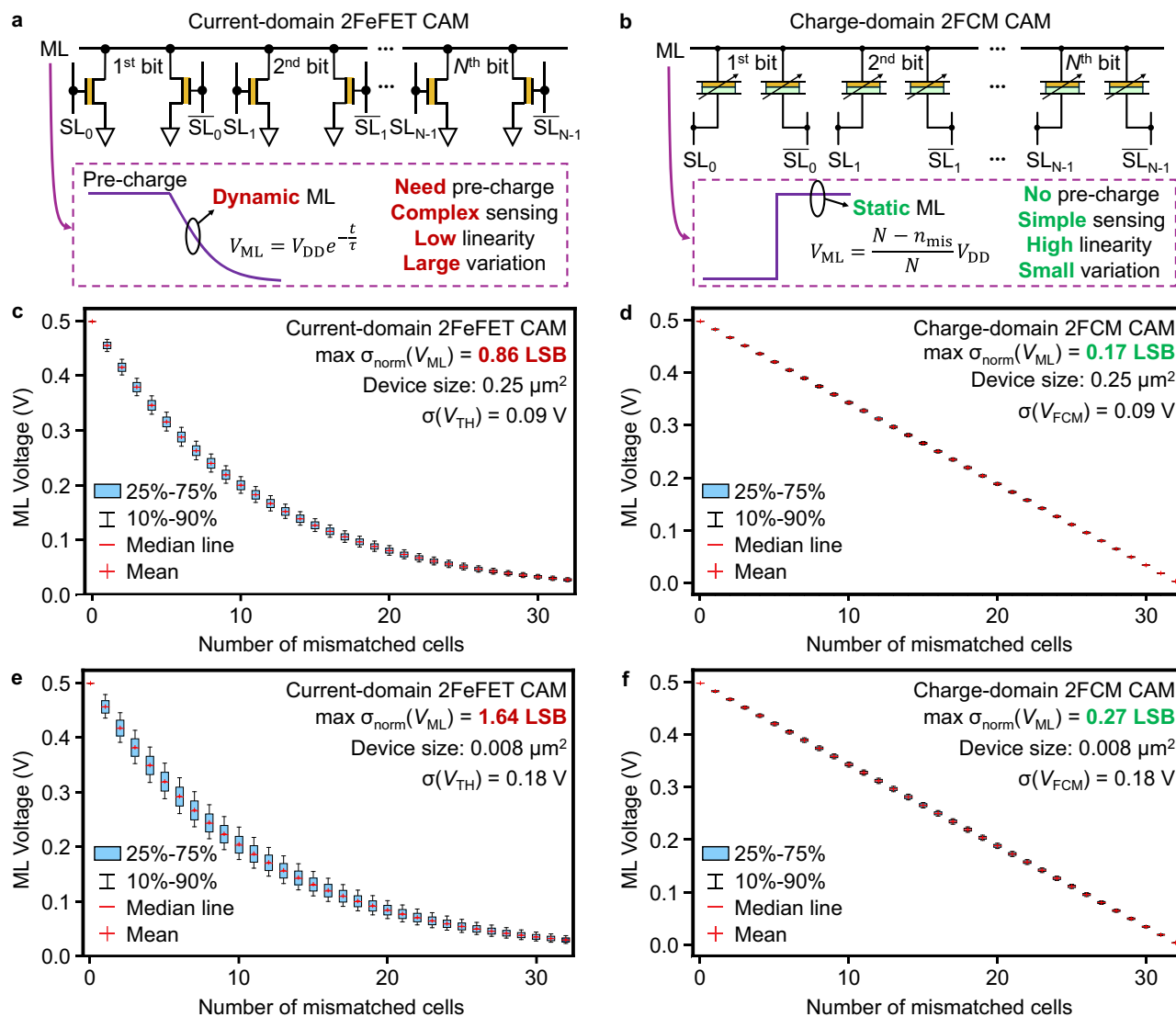


Fig. 4 | Comparison between the current-domain 2FeFET CAM and the proposed charge-domain 2FCM CAM. **a** Sensing method of the current-domain 2FeFET CAM that tracks the voltage drop of pre-charged ML⁴. This sensing method is dynamic with decreasing V_{ML} and more vulnerable to device variation. The output is not linear that leads to non-uniform sensing margin. **b** Sensing method of the proposed charge-domain 2FCM CAM. The proposed 2FCM CAM enjoys linear and static output, enabling much simpler sensing peripherals. Besides, 2FCM CAM do not require energy-hungry pre-charge operation and shows better variation

immunity. **c** V_{ML} of 32-bit 2FeFET CAM with $500 \text{ nm} \times 500 \text{ nm}$ device size and $0.09 \text{ V } \sigma(V_{TH})$ ²². **d** V_{ML} of 32-bit 2FCM CAM with $500 \text{ nm} \times 500 \text{ nm}$ device size and $0.09 \text{ V } \sigma(V_{FCM})$. **e** V_{ML} of 32-bit 2FeFET CAM with $240 \text{ nm} \times 34 \text{ nm}$ device size and $0.18 \text{ V } \sigma(V_{TH})$ ²². **f** V_{ML} of 32-bit 2FCM CAM with $240 \text{ nm} \times 34 \text{ nm}$ device size and $0.18 \text{ V } \sigma(V_{FCM})$. Box plots show the median (red line inside the box), mean (red cross inside the box), the range from 25% to 75% (box), and the range from 10% to 90% (whiskers).

Implementation of one-shot/few-shot learning on 2FCM CAM

Based on the above outstanding characteristics, one-shot/few-shot learning was implemented on the proposed 2FCM CAM as a use case to evaluate the system-level performance. The overall dataflow is illustrated in Fig. 5a. Here we use the prototypical network³⁹ (Supplementary Fig. 25) as the base model and the Omniglot dataset⁴⁰ for one-shot/few-shot learning applications. Similar to the method in the prior work⁴, we replace the fully connected layer in the MANN with an LSH function layer. During the training phase of an M -way K -shot task, LSH generates an $M \times N$ binary matrix, where N is the number of hashing bits defined by the LSH function. The pre-trained $M \times N$ binary matrix can be stored in one or multiple 2FCM CAM arrays determined by both the matrix size and the CAM array size. During the inference phase, the input query feature vector is firstly binarized after the feature extraction and LSH function, which is then compared with all vectors stored in the 2FCM CAM arrays in a parallel fashion. By finding the vector best matching the query

vector with the minimum HD, the corresponding class can be output as the final classification result.

In the experiment, the inference of a 16-way learning task with 16 hashing bits was demonstrated. After writing the 16×16 binary matrix generated by LSH into the 2FCM CAM array, we input a query feature vector representing a new unseen example through the SLs/SLs. Figure 5b shows the measured matching degree of each channel (obtained by normalizing the measured V_{ML} with a linear function), which is compared with the results calculated by software. It is seen that the 2FCM CAM correctly classifies the example, and the measured results are very close to their corresponding digital reference. The average error of the HD computation in this case is low to only 0.126 LSB. As shown in Fig. 5c, such a small error is able to realize only -0.7% (from 97.76% to 97.08%) and -0.6% (from 98.10% to 97.52%) accuracy loss in the 5-way 1-shot and 5-way 5-shot learning, respectively, for a 2FCM CAM array over software LSH with 256 hashing bits. Both values are also close to the accuracies obtained by GPU and Euclidean distance (98.4% and 99.5%).

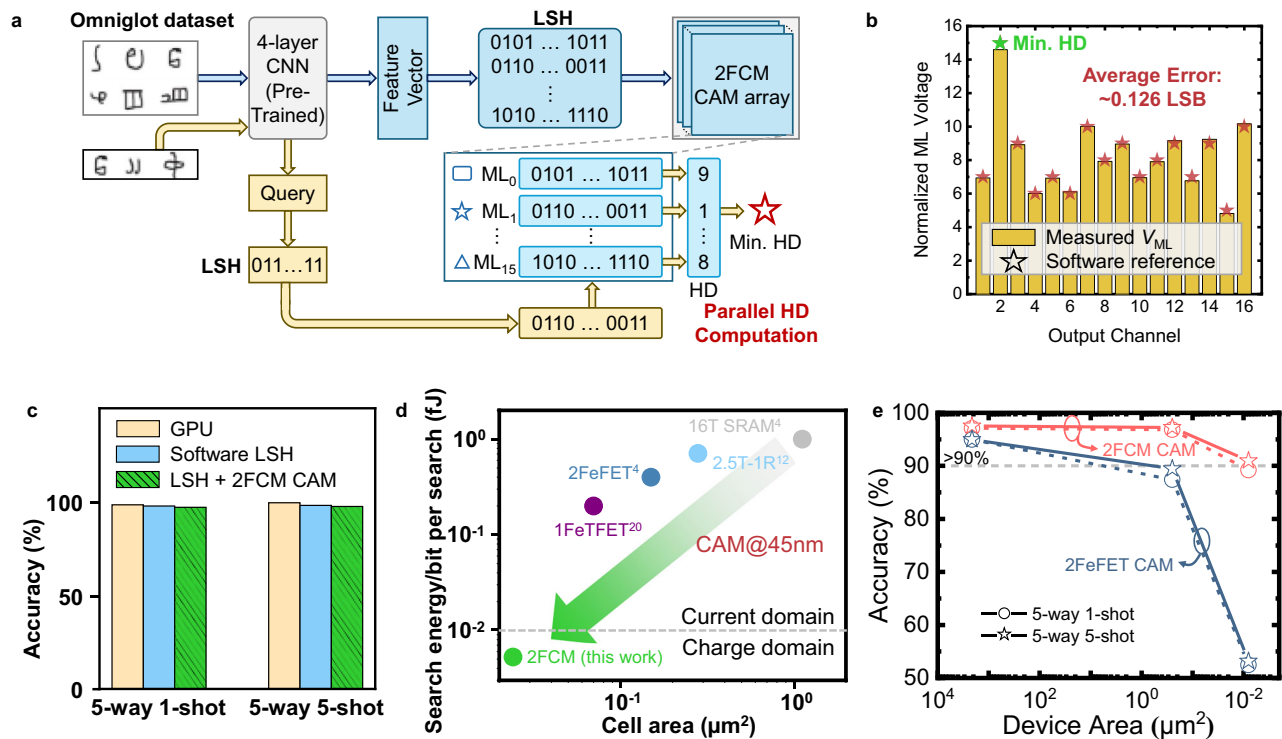


Fig. 5 | Implementation of one-shot/few-shot learning on the proposed 2FCM CAM array. **a** Dataflow of one-shot/few-shot learning implemented on our FCM CAM array. **b** A case of finding the most matched class with minimum HD for a new example computed by the fabricated 16×16 CAM array. The data is from the Omniglot dataset, and the feature vector of the most matched class is stored in Channel #2. It is seen that the degree of mismatching based on the measured V_{ML} is very close to digital HD reference with the average error of only 0.126 LSB for the 16-level output. **c** Accuracy of one-shot/few-shot learning on the Omniglot dataset using GPU, software LSH, and LSH + 2FCM CAM. For LSH + 2FCM CAM, the measured 0.126-LSB error from (b) has been added during the inference process. The

proposed 2FCM CAM array can achieve 97.1% and 97.5% accuracy (with only 0.7% and 0.6% accuracy loss compared with software LSH) on the 5-way 1-shot and 5-way 5-shot tasks, respectively. **d** Comparison of the proposed 2FCM CAM and existing CAMs for one-shot/few-shot learning in 45 nm process node. The FCM characteristics in 45 nm node are estimated based on the measured capacitance density in Fig. 2d. The proposed 2FCM CAM can achieve a low search energy of 0.005 fJ/bit for each search operation within $0.024 \mu\text{m}^2$ cell area. **e** Accuracy comparison between the proposed 2FCM CAM and the current-domain 2FeFET CAM under different device sizes. 2FCM CAM exhibits better variation immunity and can maintain >90% accuracy even under severe device variation²² for the highly scaled planar cells.

In addition to the high accuracy enabled by the charge-domain computing, the proposed 2FCM CAM structure also significantly improves the memory density and search energy efficiency of NV-CAMs compared with the current-domain CAMs in the same technology node. Figure 5d summarizes the search energy consumptions and cell areas of different CAMs in the 45 nm node. The proposed 2FCM CAM can achieve search energy as low as 0.005 fJ/bit (without peripherals) for each search operation with $0.024 \mu\text{m}^2$ cell area (Supplementary Figs. 26 and 27). Such a low search energy is mainly achieved by exploiting the energy-efficient charge-domain computing^{19,33}, and can be reduced even further with a lower V_{DD} or the energy recovery technique³¹. For write operations, the 2FCM CAM can achieve low write power comparable to that of the 2FeFET CAM⁴, owing to the similar structures between FeFET and FCM. The field-driven write mechanism inherent to ferroelectric memories ensures substantially lower write power consumption compared with resistive memories relying on current-driven write mechanisms²¹.

With the superior variation immunity compared to current-domain CAMs, the accuracy improvement offered by 2FCM CAM becomes more prominent when the device variation increases during the device down-scaling. As predicted in Fig. 5e, the proposed charge-domain 2FCM CAM can maintain high accuracy in both 5-way 1-shot (89.18%) and 5-way 5-shot (90.95%) learning tasks, even with $0.008 \mu\text{m}^2$ FCM devices. In contrast, the accuracy of the 2FeFET CAM declines rapidly as the device size decreases, raising concerns about the scalability of this CAM structure.

Furthermore, based on the above results and analysis, we evaluate the 2FCM CAM and compare it with several existing CAM designs at the system level through SPICE simulations performed on the 256×128 CAM array with peripherals in 45 nm process node (Supplementary Fig. 28). Table 1 summarizes the performance of the proposed 2FCM CAM and the CAMs reported in prior works^{4,12,13,18,20}. The proposed 2FCM CAM achieves 19.1 Mb/mm^2 memory density and only 0.74 pJ search energy for the whole 256×128 array in each search operation, making it the most promising candidate among the various NV-CAMs for the low-power and accurate HD computation with long data vectors.

Discussion

We propose and demonstrate the first charge-domain non-volatile 2FCM CAM based on the high-performance inversion-type FCM. By storing data as the device capacitance, the innovative CAM operates in the charge domain, which revolutionizes the current-domain mechanisms of conventional NV-CAMs and offers multiple remarkable advantages. We fabricated and tested a 16×16 2FCM CAM array integrated on the SOI platform. By leveraging the high capacitance ratio of the inversion-type FCM, we demonstrate that the 2FCM CAM can directly output the HD as stable voltage levels with a large sensing margin. The charge-domain operation further enables an ultra-high linearity of the output and effectively mitigates the impact of device variations. 16-channel parallel HD computation with 16-bit data in each channel was experimentally demonstrated with a low measured error, which can provide a high accuracy of 97.5% for one-shot/few-shot

Table 1 | Comparison between the proposed 2FCM CAM and prior CAM designs

Reference	NE 2019 ^a	VLSI 2021 ¹³	ISSCC 2016 ¹²	NE 2019 ^a	IEDM 2021 ¹⁸	VLSI 2022 ²⁰	This work
CAM cell	16T SRAM	2T-2RRAM	2.5T-1RRAM	2FeFET (planar)	2FeFET (Monolithic 3D)	1FeTFET	2FCM ^d
Demonstration complexity (# of CAM cells)	N/A	256 × 128	64 × 128	1 × 6	10 × 10 × 2 (2 layers)	Single device	16 × 16
CAM domain	Current	Current	Current	Current	Current	Current	Charge
Technology node (nm)	45 ^a	40	45 ^a	45 ^a	45 ^a	45 ^a	45 ^a
Evaluated array size	256 × 128 ^a	256 × 128	256 × 128 ^a	64 × 64 × 8 ^a	256 × 128 ^a	256 × 128 ^a	256 × 128 ^a
Cell area (μm ²)	1.12	0.22	0.28	0.15	0.15	0.07	0.024
Memory density (Mb/mm ²)	0.8	0.3 ^b	3.1	5.4	18.2	10.0	19.1
Search delay ^c (ns per search) (w/ peripherals)	1.0	5.0 ^b	1.0	1.4	1.1	1.2	0.75
Search energy (fJ per bit per search) ^c (w/o peripherals)	1.0	N/A	0.7	0.4	N/A	0.2	0.005
Search energy (pJ per search) ^c (w/ peripherals)	33.3	270 ^b	23.8	13.7	2.1	7.1	0.74

^aThe setting for SPICE simulation.^bChip-level results.^cAll results are evaluated under 1 V search voltage.^dThe first FCM-based CAM known to us.

learning tasks. In addition, our simulation reveals that the charge-domain search operation of the proposed 2FCM CAM brings the energy efficiency of NV-CAM to the next level with a high integration density. This innovative CAM design outperforms the existing NV-CAMs in various aspects, manifesting tremendous potential as a technology enabler and game changer to realize reliable and energy-efficient in-memory search for future low-power edge-computing applications.

Methods

Device fabrication

In this work, the inversion-type FCM devices were fabricated on a SOI substrate with HAO ferroelectric thin film. The key process steps for the device fabrication are listed in Supplementary Fig. 1a. The SOI wafer has a -220 nm p⁺ silicon layer with a resistivity ≥750 Ω·cm. Before the process, the wafer was cleaned by dilute hydrofluoric acid to remove the native oxide on the silicon surface. The -10 nm HAO film was grown by thermal atomic layer deposition (ALD) under 300 °C deposition temperature. Tetrakis(ethylmethyldamido)hafnium (TEMAHf) and Trimethylaluminum (TMA) were used as the precursors for Hf and Al, respectively. The oxidant was ozone (O₃). The concentration of Al in the film was controlled by setting the Al to Hf cycle ratio of the ALD process, which was optimized to 1:30 to achieve good ferroelectricity. The HAO layer was then covered by a -50 nm W layer deposited by physical vapor deposition (PVD). After patterning and etching the TEs, self-aligned phosphorus implantation was performed to introduce the n⁺ region. The devices were annealed at 650 °C for 30 s by the rapid thermal annealing process. After that, these inversion-type FCM devices were isolated by etching through the silicon layer. Next, a -100 nm Al₂O₃ dielectric layer was deposited by ALD followed by patterning and etching to open the VIAs. Lastly, W metal lines were formed by PVD to connect the devices according to the proposed 2FCM CAM array structure.

Electrical characterization

The electrical characterization of the inversion-type FCM devices was performed in dark environment and room temperature by Keithley 4200-SCS semiconductor analyzer and aixACCT TF Analyzer 3000. The C-V measurement was performed by 4200-CVU (capacitance

voltage unit) in the semiconductor analyzer under a small signal with the root-mean-square voltage of 50 mV and 100 kHz frequency. Two 4225-PMUs (pulse measurement units) were used to write the devices with ±5.5 V 10 μs voltage pulses. Note that the relatively large pulse amplitude and long pulse width were applied to ensure that the devices were written to their saturation states, which can be further optimized. The retention and endurance measurement were conducted by TF Analyzer 3000.

The measurement and demonstration of the 2FCM CAM were performed on a customized test platform (the details can be found in Supplementary Fig. 11). In the platform, STM32H7 microcontroller was programmed to control the peripherals to drive the array, which communicated with an upper computer to implement the CAM operation and learning tasks. Besides, Tektronix MDO3104 oscilloscope was utilized to record the ML voltage waveforms for demonstration.

Data availability

The data that support the figures within this paper and other findings of this research are available in the Figshare repository at <https://doi.org/10.6084/m9.figshare.29816003>.

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Author contributions

X.G. and X.L. proposed and supervised the research. X.L. and H.Z. conceived the FCM-based CAM design with corresponding operations. Z.Z. [Zuopu Zhou] performed the FCM-based array fabrication and the

experimental verification. H.Z. performed the circuit-level simulation and task-level evaluation. L.J., Z.Z. [Zijie Zheng], H.Y, T.K. and K.N. contributed to the discussion and data analysis. Z.Z. [Zuopu Zhou], H.Z., X.L. and X.G. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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